EDDY OGOLA

New York City, NY | eddy.ogola1@gmail.com | +1-617-251-4816 | linkedin.com/in/eddy-ogola | eddogola.github.io

EDUCATION

Massachusetts Institute of Technology **May 2025**

BSc. Computer Science | Artificial Intelligence & Decision Making *Cambridge, MA*

- Relevant Classes: *Graduate-level Machine Learning, TinyML & Efficient Deep Learning Computing, Optimization Methods*

RELEVANT EXPERIENCE

Meta **May - August 2023 | July 2025 - May 2026 | May 2024 - August 2024**

Software Engineer *New York City, NY*

AI Infrastructure | Pytorch Distributed Team

- Spearheaded **distributed checkpointing** for large training workloads utilizing the **pipeline parallel distributed paradigm** in multi-GPU environments, resulting in at least a **10x reduction in training downtime**.
- **Streamlined legacy modules within `torch.distributed`** improving code maintainability and reducing technical debt in preparation for integration into the main `torch` codebase.
- **Architected** modular, **extensible interfaces for distributed state management**, decoupling checkpointing logic from specific hardware topologies to support future scaling requirements **across 3 distributed paradigms**: data, pipeline & tensor parallel.

Infrastructure Security Engineering | Zero Trust Networking Team

- Architected and automated production secure-channel onboarding using an **AI-driven agent**, reducing cross-team dependency cycles from **2 weeks to under an hour** while eliminating **90% of manual compliance checks**.
- Engineered an internal **security policy testing CLI tool and CI pipeline signal** that replaced a 6-hour production canary process with a **1-minute local check**, accelerating deployment velocity by **60%** and cutting compute overhead.
- **Led the zero-downtime migration** of a core policy library out of an edge proxy serving **6M req/s**, reducing the edge blast radius and decoupling the architecture to enable standalone adoption across **3+ microservices**.

Monetization | Ads Serving Platform Team

- **Overhauled the testing infrastructure** for a high-traffic database API service handling **37.3M daily reads**, mitigating critical regression vectors to improve production uptime by **5%**.
- **Engineered data ingestion and aggregation pipelines** to synthesize granular compute resource metrics into high-level MegaWatt cost indicators, unifying observability across **50+ teams** into a single analytics dashboard.

Broad Institute **September 2023 - May 2024**

Eric and Wendy Schmidt Center Funded Research and Innovation Scholar *Cambridge, MA*

- **Ported and optimized deep learning inference engines** for biological image segmentation models to the browser using ONNX Runtime, enabling low-latency, client-side execution achieving a **5s** end-to-end execution time
- **Adapted YOLACT & U-Net architectures** by swapping back-end encoders & strategically freezing layers, optimizing real-time instance segmentation for microbiological images, reducing the YOLACT model from **~50M to ~25M parameters**.

MIT CSAIL **February - May 2022**

Undergraduate researcher *Cambridge, MA*

- Researched implementation of **time-series data imputation frameworks(e.g. MICE, Dynamic Bayesian Networks, & recurrent GANs)** to reconstruct highly fragmented clinical streams with up to **25% missing values**, restoring data integrity for downstream analytics.
- Engineered & benchmarked **customized Generative Adversarial Networks (GANs) to simulate complex time-series medical telemetry (ECG/EHR)**, mitigating data scarcity while maintaining strong statistical alignment with empirical patient cohorts.

PROJECTS

samaki.ai **March 2026**

- Engineered an **automated aquaculture platform** for the **MIT Africa Innovation Conference Hackathon**, deploying a **custom YOLOv12 object detection pipeline to track fish health metrics** and behavioral patterns in real-time.
- Designed and **prototyped custom feeder hardware integrated directly with the computer vision engine**, enabling automated, closed-loop feeding cycles driven entirely by live visual telemetry.

TECHNICAL SKILLS: Python, C++, Go, PyTorch, LoRA fine-tuning, Claude code, RAG, Distributed Systems, CAD, Tmux, Nvim